

Telemedicine System

Project Report



Submitted To:
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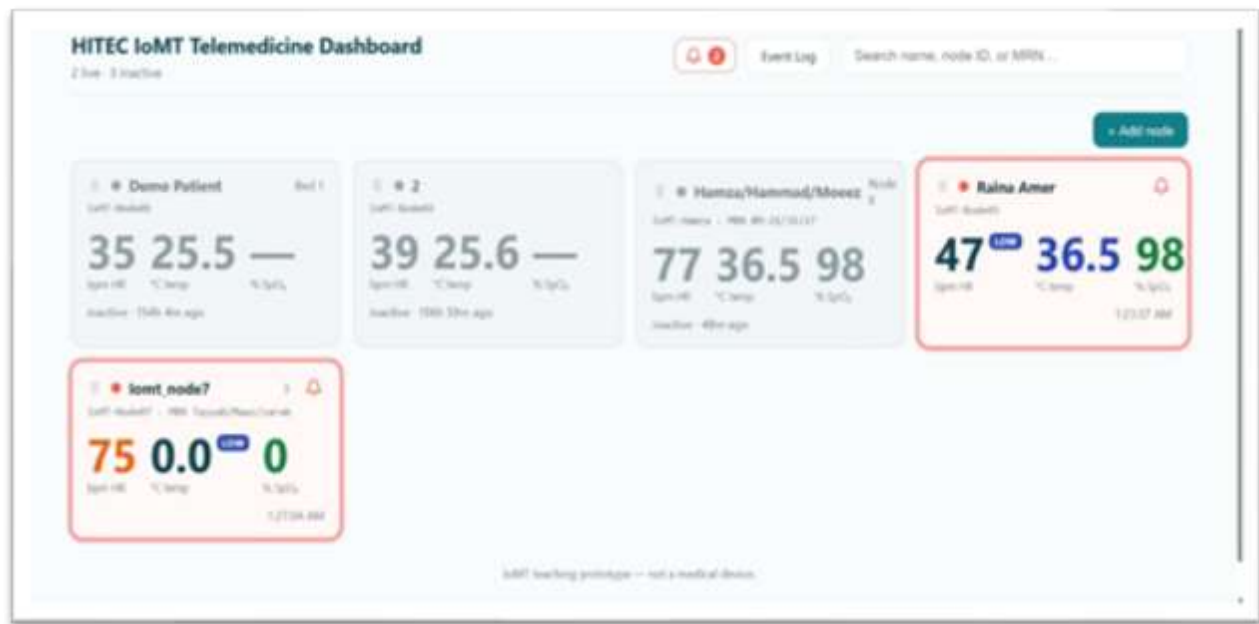
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IoMT Telemedicine Dashboard

Real-Time Vital Signs Monitoring using ESP32 & MAX30102

Sensor Used	MAX30102 (Heart Rate, Temp & SpO ₂)
Microcontroller	ESP32 (Wi-Fi enabled)
Platform	HITEC IoMT Telemedicine Dashboard (IoMT-Node05)



1. Summary

This project presents a low-cost, IoMT-based telemedicine system designed to monitor real-time vital signs specifically Heart Rate (HR) and Oxygen Saturation (SpO₂) using the MAX30102 pulse oximetry sensor interfaced with an ESP32 Wi-Fi microcontroller. The data are transmitted wirelessly to a centralized web-based telemedicine dashboard developed by the instructor (HITEC IoMT Telemedicine Dashboard), where each student group maintains an individual patient node. Our group implemented IoMT-Node05, registered under the patient's name "Raina Amer," and successfully streamed live physiological data visible to the entire class. The system demonstrates the core workflow of an Internet of Medical Things (IoMT) architecture: sensor acquisition, embedded firmware processing, Wi-Fi transmission, and cloud-based clinical visualization illustrating how remote patient monitoring (RPM) can be practically realized using affordable off-the-shelf hardware.

2. Introduction

IoMT refers to a connected ecosystem of medical devices that collect and transmit patient data remotely, enabling continuous monitoring beyond clinical settings. Heart rate (HR) and SpO₂ are critical vital signs, low SpO₂ (<95%) indicates hypoxemia while abnormal HR signals arrhythmia. The MAX30102 sensor uses photoplethysmography (PPG) with red (660nm) and infrared (880nm) LEDs to non-invasively compute both parameters via I2C interface, making it ideal for low-cost IoT integration. The ESP32 microcontroller, with built-in Wi-Fi, acquires PPG data, processes HR and SpO₂ on-chip, and transmits results to a remote server via HTTP.

Literature supports this approach: Siddiqui et al. (2021) validated ESP32-based real-time patient monitoring; Maheswari & Anuradha (2022) confirmed MAX30102 accuracy for wearable SpO₂ measurement. Existing systems, however, are standalone prototypes without multi-node institutional integration. This project extends prior work by connecting a real sensor node to the HITEC IoMT Telemedicine Dashboard, a multi-node platform, demonstrating a scalable, classroom-level RPM architecture using entirely off-the-shelf components.

3. Objectives

- ❖ Interface the MAX30102 sensor with an ESP32 to acquire real-time Heart Rate (bpm) and SpO₂ (%) data using photoplethysmography.
- ❖ Transmit acquired vital signs wirelessly over Wi-Fi to the HITEC IoMT Telemedicine Dashboard and maintain a live patient node (IoMT-Node05).
- ❖ Demonstrate an end-to-end IoMT pipeline from sensor acquisition and embedded firmware to cloud visualization as a proof-of-concept for low-cost remote patient monitoring.

4. Scope

The scope of this project is limited to the hardware interfacing of the MAX30102 sensor with the ESP32 microcontroller, firmware development for real-time HR and SpO₂ extraction, and Wi-Fi-based data upload to the instructor-provided HITEC IoMT dashboard. The project does not include clinical-grade accuracy validation, or multi-parameter monitoring beyond HR and SpO₂. It serves as an educational prototype to demonstrate core IoMT concepts within a controlled university lab environment.

5. Methodology

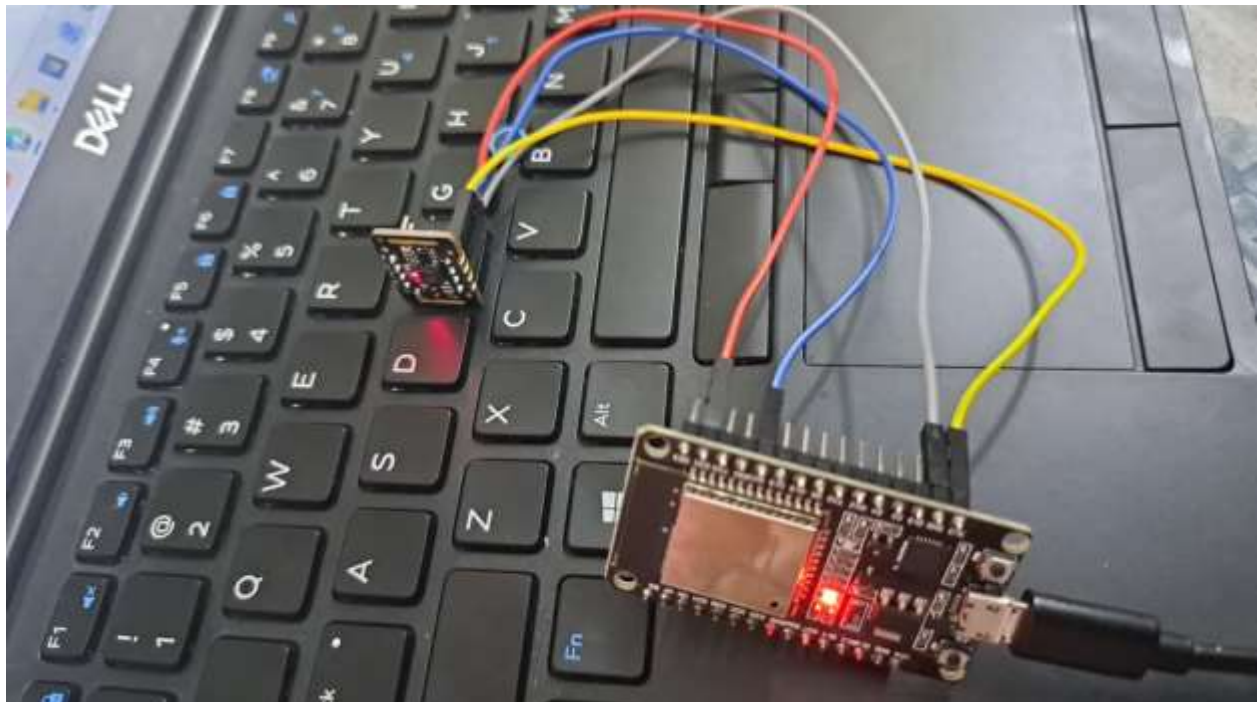
5.1 Hardware Setup

The following components were used:

- ESP32 Development Board (Wi-Fi + dual-core MCU)
- MAX30102 Heart Rate & SpO₂ Sensor Module
- Jumper wires, USB cable for power/programming

Wiring connections (I2C):

MAX30102 Pin	ESP32 Pin
VIN	3.3V
GND	GND
SDA	GPIO 21 (default I2C SDA)
SCL	GPIO 22 (default I2C SCL)

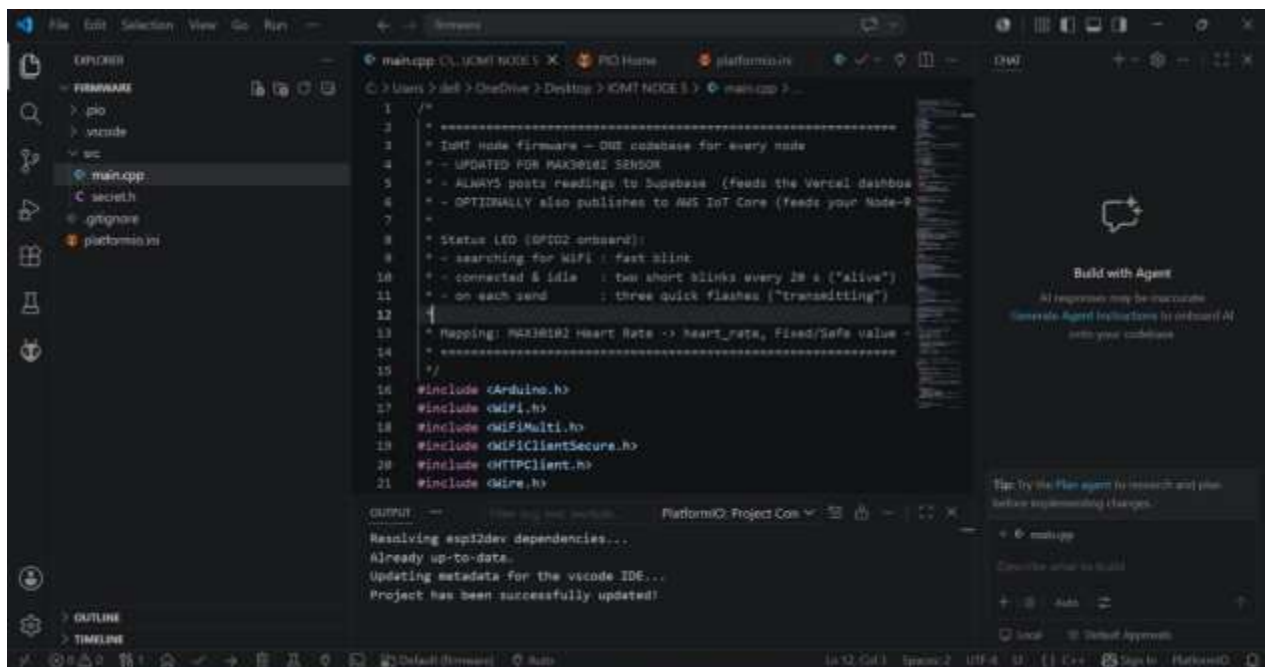


5.2 Firmware Development (VS Code + PlatformIO + ESP32)

The firmware was developed using **VS Code with the PlatformIO IDE extension** with the ESP32 configured as the target board (esp32dev). The project structure included src/main.cpp as the main firmware file, secret.h for storing Wi-Fi credentials and API keys separately, and platformio.ini for board and library configuration. Key firmware steps:

- ❖ Included libraries: Arduino.h, WiFi.h, WiFiMulti.h, WiFiClientSecure.h, HTTPClient.h, Wire.h, and the MAX30102 sensor library.
- ❖ Configured the onboard Status LED (GPIO2) with three behavioral modes: fast blink during Wi-Fi search, two short blinks every 20s when connected and idle, and three quick flashes on each data transmission.
- ❖ Initialized I2C communication with the MAX30102 and configured sensor parameters (LED brightness, sample rate, pulse width).
- ❖ Implemented real-time loop to read raw Red and IR PPG values and extract Heart Rate (bpm) and SpO₂ (%) using the on-chip algorithm.
- ❖ Data was posted on dashboard.

Initial firmware only transmitted Heart Rate. The code was subsequently updated to include SpO₂ in the payload after observing missing values on the dashboard.



5.3 Dashboard Node Registration & Data Upload

The instructor pre-configured the HITEC IoMT Telemedicine Dashboard with student nodes. Our node (IoMT-Node05 / Raina Amer) was initially deleted inadvertently; it was subsequently re-registered and reconnected. Once the updated firmware was flashed and the sensor was placed on a fingertip, data began appearing live on the dashboard, showing HR, temperature, and SpO₂ values with a timestamp.

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OUTPUT  ...  Filter (e.g. text, !exclude...)  PlatformIO: Project Con  x  🔒  ...  [ ]  x

[Resolving] esp32dev dependencies...
Already up-to-date.
Updating metadata for the vscode IDE...
Project has been successfully updated!

Default (firmware)  🔌  Auto  Ln 12, Col 3
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  +  v  ...  [ ]  x

*  Executing task: C:\Users\dell\.platformio\env\Scripts\platformio.exe device monitor
=== IoT node: IoT-Node05 ===
Connecting WiFi...

WiFi OK: "Home"  IP=192.168.1.19
[Sensor] Finger detected. Live Avg Heart Rate: 94.0 BPM
[Supabase] {"device_id":"IoT-Node05","body_temp":36.5,"heart_rate":94.0} -> HTTP 201
[185369][E][Wire.cpp:499] requestFrom(): i2cWriteReadNonStop retu

Default (firmware)  🔌  Auto  Ln 18, Col 21
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Build Task ✓

Clean Ta... ✓

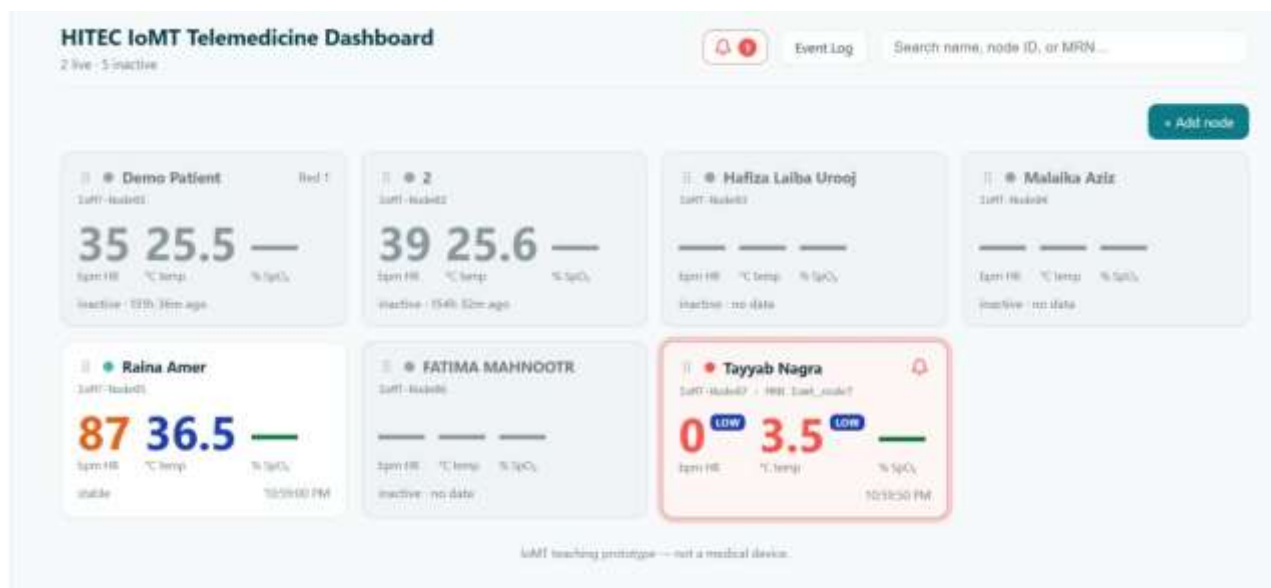
Clean (e... ✓

Upload... ✓

Upload (... ✗

Monitor... ↻

PlatformIO ...



5.4 Challenges Encountered

- ❖ Node deletion: Our node was accidentally removed from the dashboard mid-session; the node was recreated and the ESP32 was re-flashed to restore connectivity.
- ❖ Initial SpO₂ missing: The first firmware version did not include SpO₂ in the payload. After identifying the gap, the code was updated and the value was successfully streamed.
- ❖ Sensor failure at end of session: The MAX30102 stopped functioning toward the end of the experiment, causing HR and SpO₂ to read 0. Screenshots of valid readings were captured earlier and are included in this report.
- ❖ HR flagged LOW: The dashboard flagged Heart Rate as LOW (47 bpm) with a blue LOW badge. This may be due to finger placement or transient signal artifact at time of capture.

6. Expected Results and Benefits

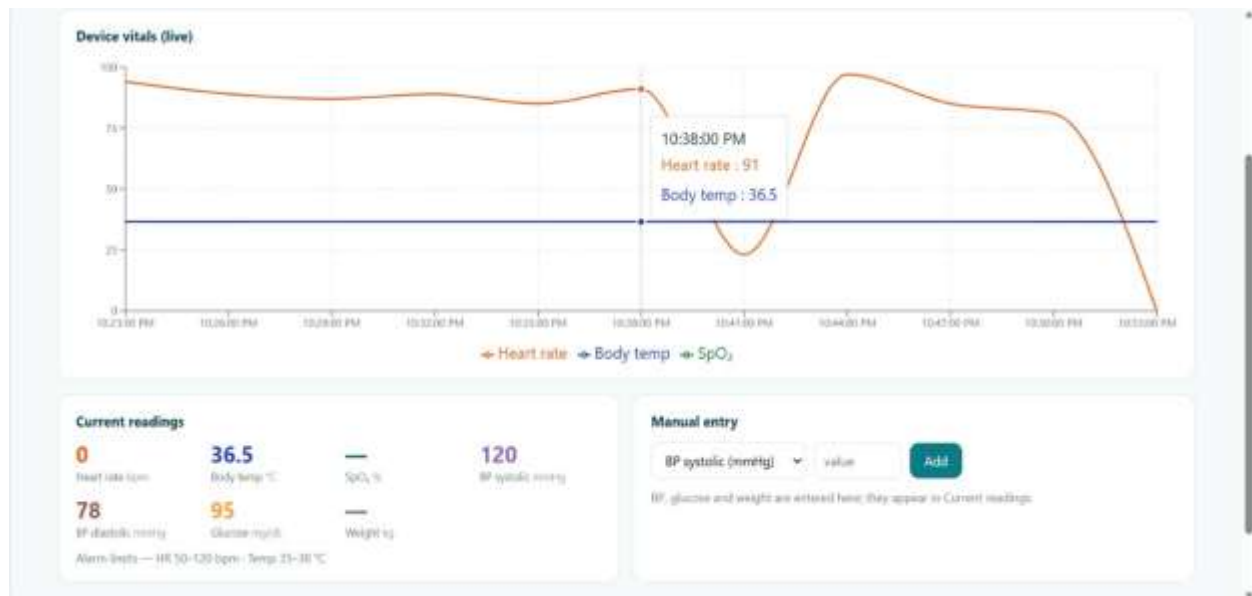
The system successfully demonstrated real-time transmission of physiological data from a wearable MAX30102 sensor to a centralized web dashboard over Wi-Fi. Valid readings captured during the live session included a Heart Rate of 47 bpm (flagged LOW), body temperature of 36.5°C, and SpO₂ of 98% all displayed on the dashboard with a timestamp of 1:23:37 AM on IoMT-Node05.

Expected benefits of this IoMT architecture include:

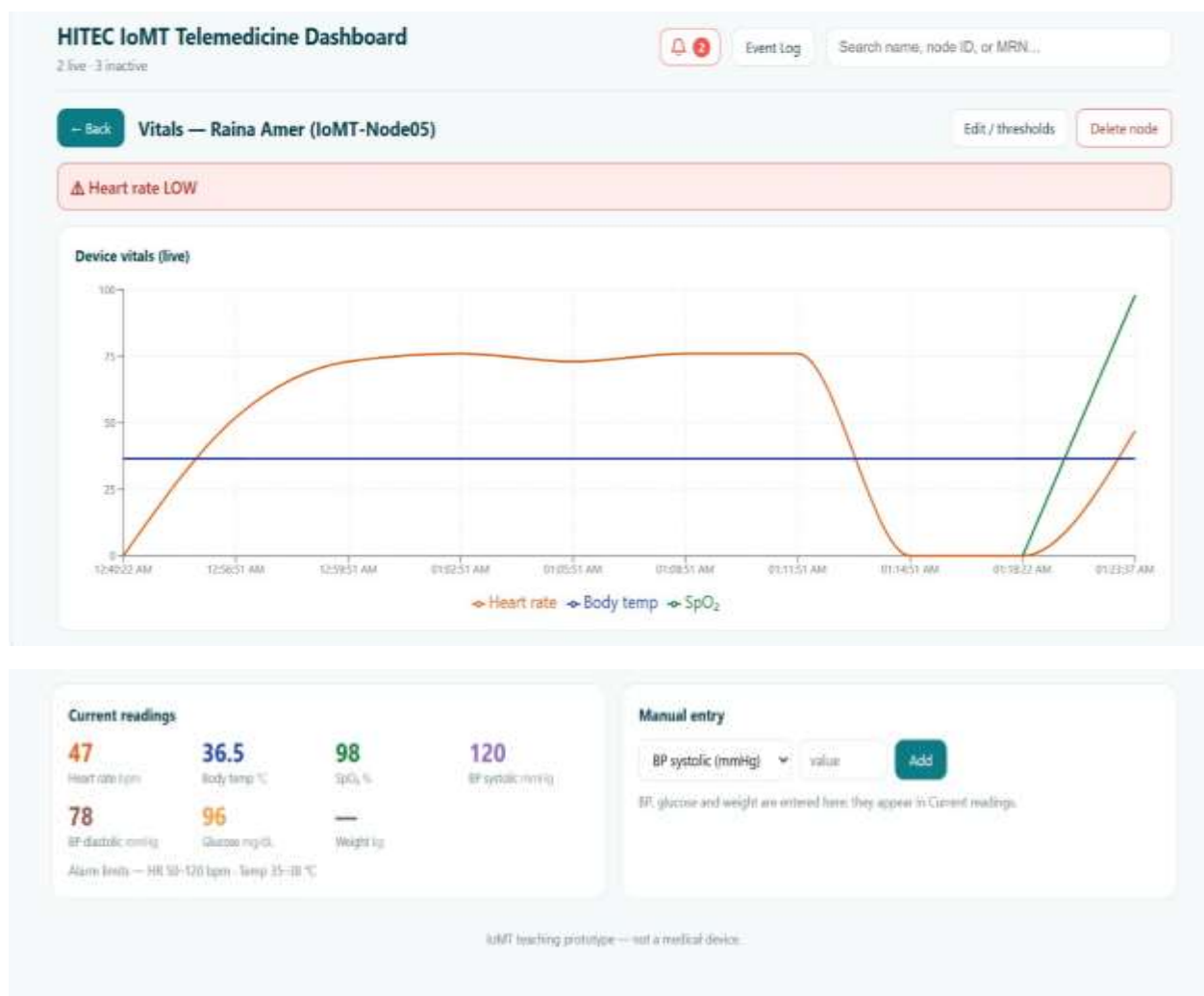
- ❖ Continuous remote monitoring of at-risk patients without requiring hospital visits.
- ❖ Early detection of hypoxemia (low SpO₂) or bradycardia (low HR) with automated alerting.
- ❖ Cost-effective implementation using commercial off-the-shelf (COTS) components.
- ❖ Scalable multi-node architecture suitable for ward-level or community-level monitoring.

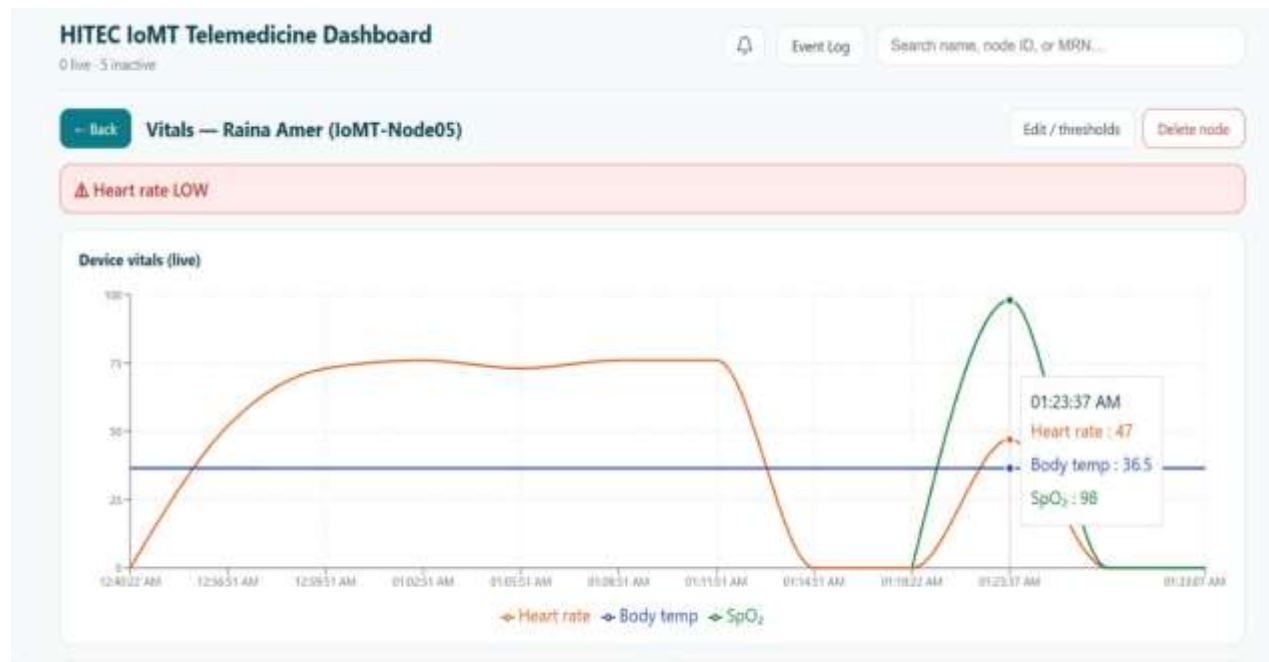
Initial stage:





Data Apperance again after node deletion was handled:





7. Key Outcomes of the Work Proposed

- ❖ **Prototype:** A functional IoMT node built from ESP32 + MAX30102, capable of real-time vital sign acquisition and Wi-Fi transmission to a remote dashboard.
- ❖ **Technology:** Demonstrated practical use of I2C sensor interfacing, PPG-based SpO₂/HR computation, and HTTP data upload in an embedded health monitoring context.
- ❖ **Process:** Established a repeatable node setup workflow firmware flashing, network configuration, node registration, and live data verification applicable to larger deployments.
- ❖ **Innovation:** Extended the instructor's dummy-data dashboard to carry actual physiological sensor data, validating the dashboard's architecture for real clinical prototype use.
- ❖ **Commercialization potential:** The ESP32 + MAX30102 stack can be packaged as a low-cost wearable patch for RPM in resource-limited healthcare settings (e.g., rural clinics, home isolation during pandemics).